Garvit Singh

Backend Developer | Node.js, PostgreSQL & AI Integration

Jaipur, Rajasthan, India | garvitsingh171@gmail.com | +91 9690067289
LinkedIn: garvitsingh171 | GitHub: garvitsingh171 | Portfolio: garvitsingh171.com

PROFESSIONAL SUMMARY

Second-year Computer Science undergraduate experienced in taking product requirements from design through testing and release. Brings careful debugging, early blocker communication, and a collaborative approach to code reviews while maintaining ownership of deliverables.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, SQL, Python, Java
Backend & APIs: Node.js, Express.js, REST API Development, Authentication, Authorization, RBAC, JWT, Zod
Databases & ORM: PostgreSQL, Prisma ORM, SQL Querying, Relational Schema Design, Database Migrations
Testing & Tools: Vitest, Unit/API Testing, Git, GitHub, Code Reviews, Postman, Swagger/OpenAPI, Docker, Linux
AI Integration: Gemini API, Retrieval-Augmented Generation (RAG), Embeddings, pgvector, Vector Search
Additional: Next.js, React.js, MongoDB, Supabase, Pino

PROJECTS

Pravaah — Clinic Operations & Appointment Management Jun 2026 – Present
Live Demo | GitHub | Node.js, Express.js, TypeScript, PostgreSQL, Prisma, Clerk, Zod, Vitest

- Developed and documented **39 REST API endpoints** across 9 Express.js modules; designed PostgreSQL schemas with **12 Prisma models** and **20 database migrations** for appointments, queues, patients, and doctors.
- Implemented Clerk authentication, Admin/Staff role-based access control (RBAC), clinic-scoped authorization, and Zod input validation; verified cross-clinic request rejection in release checks.
- Coordinated appointment and queue updates with **14 Prisma transaction blocks** and **7 PostgreSQL advisory locks**, including doctor schedule-conflict checks and concurrent-write safeguards.
- Wrote **380+ automated unit and API tests** across **65 Vitest files**, covering authorization, appointment and queue logic, request validation, and API error handling.

ResolveAI — RAG-Powered Support Issue Resolution Sep 2026 – Present
GitHub | Next.js API Routes, TypeScript, PostgreSQL, pgvector, Gemini API, Zod

- Built an AI-assisted backend API with Zod validation, Gemini embeddings, PostgreSQL/pgvector **top-4 retrieval**, grounded RAG responses, server-only credentials, and retries for transient errors.

EXPERIENCE

Mathesar Foundation — Open Source Contributor Nov 2025 – Jan 2026

- Delivered **5 merged pull requests** (4 Mathesar, 1 tldr-pages) in a 9-week program, including fixes for timezone-aware data display and missing-record error states.
- Completed **23/23 sessions**; debugged issues and iterated on TypeScript/Python changes through maintainer code reviews and pull-request feedback.

ClearTax GST Reconciliation — Backend Developer (Kalvium Simulated Work) Aug 2026 – Sep 2026

- Developed authenticated Next.js route handlers and a **6-model Prisma/PostgreSQL** schema to reconcile GSTR-2B JSON with purchase-register CSV files up to **10 MB / 10,000 rows**.
- Implemented **12 validation error codes**, cursor-based pagination, centralized API error handling, and Pino logging; authored **59 of 115** team commits.

EDUCATION

JECRC University, Jaipur — B.Tech, Computer Science Engineering 2025 – 2029 (Expected)
Kalvium Software Product Engineering Program | **CGPA: 9.23/10**

ACHIEVEMENTS

- **1,600+ GitHub contributions** in the past year; **120+ LeetCode** problems solved.
- **Winner & Team Lead**, Kalvium Squad Portfolio Competition (2025); Smart India Hackathon participant (2025).
- **3rd place**, GDG × JECRC "Unlocking Agentic AI" workshop quiz (Oct 2026).